

Catherine Luse Software Engineer

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Professional Summary

Software engineer with four years of production experience building reliable web applications with Vue, React, Node.js, TypeScript, and JavaScript across startup, enterprise, open-source, and medical software environments. I specialize in maintainable frontend development, automated testing, and backend integration, with recent hands-on experience in Vue 3, Pinia, Vitest, Playwright, AWS Lambda, and Node.js services. I'm pragmatic, quality-focused, and comfortable using AI-assisted development tools to improve productivity while preserving engineering standards.

Experience

Software Engineer | HeartFlow, Remote Feb 2025 - Present

Built critical features for cardiovascular imaging software used in surgical planning, spanning Vue 3 frontend development and Node.js backend integration. Developed TypeScript components with the Composition API, Pinia, modular composables, Vitest unit tests, and Playwright coverage. Contributed to production Node.js/AWS Lambda services, including an Express-based web server for authentication, secure clinical-data proxying, and protected asset delivery. Added Datadog APM (Application Performance Monitoring) to an HMAC generation Lambda and helped expose plaque disease-stage data to the frontend. Worked in a medical-grade software environment focused on reliability, documentation, comprehensive testing, and safe deployment. **Tech stack: Vue 3, TypeScript, Pinia, Vitest, Playwright, Node.js, Express, AWS Lambda, AWS CDK, S3, CloudFront, Datadog.**

Software Engineer | Flightcrew, Remote Jan 2023 - Oct 2024

Built core features for an early-stage DevOps automation platform that helped teams evaluate and apply Kubernetes configuration recommendations. Implemented a GitOps diff view showing users exactly what infrastructure configuration changes would be made before applying them, helping build trust in automated DevOps workflows. Built user onboarding flows, permission-management features, and data visualizations showing recommendation impact. Collaborated closely with founders and backend engineers, including proposing GraphQL schema changes and implementing frontend work in parallel with backend API development. Shipped quickly in a fast-moving startup environment while maintaining code quality and product reliability. **Tech stack: TypeScript, JavaScript, React, Next.js, GraphQL, Victory Charts/D3, Kubernetes, GitOps.**

Software Engineer | SUSE, Remote Sep 2021 - Jan 2023

Contributed to Rancher, an open-source Kubernetes management platform used by over 30,000 teams globally. Built frontend features for complex enterprise Kubernetes workflows, including multi-tenancy scenarios and selective monitoring deployment across namespace subsets. Worked in a large Vue/TypeScript codebase, collaborating across engineering teams to support cluster management, observability, and configuration workflows. **Tech stack: JavaScript, TypeScript, Vue, Kubernetes, Docker.**

Education

Galvanize Certificate, Full-stack Web Development 2018

Skills

- **Frontend:** Vue 3, React, TypeScript, JavaScript, HTML, CSS, Pinia, Vitest, Playwright, Cypress, GraphQL, REST APIs.
- **Backend:** Node.js, TypeScript, Express, GraphQL. Familiar with Python, Go, microservices, REST APIs, PostgreSQL, SQL.
- **Cloud & Infrastructure:** Docker, Kubernetes, AWS, AWS Lambda, AWS CDK, Google Cloud Platform, CI/CD.
- **AI-Assisted Development:** Copilot CLI, Claude, ChatGPT.

Side Projects

- **Multiforum - 2019 to present:** Personal full-stack project designing and developing a community platform combining forums, wikis, event calendars, and file sharing. Built with TypeScript, Vue 3/Nuxt 3, Neo4j, Apollo GraphQL, and Node.js. Includes Node.js-based backend tests and Playwright end-to-end tests. <https://github.com/gennitdev/multiforum-nuxt>. Hosted demo at [topical.space](#).
- **Multiforum Plugin Security Scan Service - 2026:** Python/FastAPI file-security microservice (VirusTotal + static ZIP analysis) for the Multiforum platform; Dockerized for Cloud Run with GitHub Actions CI/CD, Terraform, and pytest. <https://github.com/gennit-project/multiforum-plugin-security-scan-service>